1 I

1.1 (1)

Using the Legendre transformation $G = F + PV$, we obtain

$$\begin{aligned} dG &= dF + PdV + VdP \\ &= -SdT + VdP + \mu dN. \end{aligned}$$

1.2 (2)

Since G is an extensive quantity, while T and P are intensive variables, Euler's theorem gives

$$G = N\left(\frac{\partial G}{\partial N}\right)_{T,P} = \mu N.$$

Taking the total differential,

$$dG = \mu dN + N d\mu.$$

Comparing this with the result of (1),

$$N d\mu = -SdT + VdP.$$

Therefore, the Gibbs-Duhem relation is

$$\boxed{d\mu = -s\,dT + v\,dP},$$

where $s = S/N$ and $v = V/N$.

2 II

2.1 (3)

Since the particles are independent and the three momentum components are independent of one another, the partition function can be factorized:

$$\begin{aligned} Z &= \frac{V^N}{h^{3N}N!}\int d^3p_1 \cdots d^3p_N \exp\left(-\frac{1}{2mk_BT}\sum_{i=1}^{N} p_i^2\right) \\ &= \frac{1}{N!}\left[\frac{V}{h^3}\left(\int_{-\infty}^{+\infty} e^{-\frac{p^2}{2mk_BT}}dp\right)^3\right]^N \\ &= \frac{1}{N!}\left(\frac{V}{\lambda^3}\right)^N, \end{aligned}$$

where

$$\lambda = \frac{h}{\sqrt{2\pi m k_B T}}$$

is the thermal de Broglie wavelength.

Using Stirling's approximation,

$$\ln N! \approx N\ln N - N,$$

the Helmholtz free energy becomes

$$\boxed{F = -Nk_BT\left[\ln\left(\frac{V}{N\lambda^3}\right) + 1\right]}.$$

2.2 (4)

The chemical potential is given by

$$\begin{aligned}\mu &= \left(\frac{\partial F}{\partial N}\right)_{T,V} \\ &= k_B T \ln\left(\frac{N\lambda^3}{V}\right).\end{aligned}$$

Using the ideal gas equation $PV = Nk_BT$, we obtain

$$\boxed{\mu = k_B T \ln\left(\frac{P\lambda^3}{k_B T}\right)}.$$

2.3 (5)

The density of states is

$$D(\epsilon) = \frac{4\pi V}{h^3}\int_0^\infty \delta\left(\frac{p^2}{2m} - \epsilon\right) p^2 dp.$$

Evaluating the integral gives

$$\boxed{D(\epsilon) = \frac{4\sqrt{2}\pi V m^{3/2}}{h^3}\epsilon^{1/2}}.$$

Comparing this with $D(\epsilon) = VA\epsilon^a$,

$$\boxed{A = \frac{4\sqrt{2}\pi m^{3/2}}{h^3}, \qquad a = \frac{1}{2}}.$$

2.4 (6)

From the definition of the grand potential,

$$\Omega = F - \mu N,$$

its total differential is

$$d\Omega = -SdT - PdV - Nd\mu.$$

Therefore,

$$P = -\left(\frac{\partial \Omega}{\partial V}\right)_{T,\mu}, \qquad N = -\left(\frac{\partial \Omega}{\partial \mu}\right)_{T,V}.$$

Using the Maxwell relation,

$$\boxed{\left(\frac{\partial P}{\partial \mu}\right)_{T,V} = \left(\frac{\partial N}{\partial V}\right)_{T,\mu}}.$$

From (5), the particle number is

$$N = VA\int_0^\infty \frac{\epsilon^a}{e^{(\epsilon-\mu)/k_BT} - 1} d\epsilon.$$

Thus,

$$\frac{\partial P}{\partial \mu} = A \int_0^\infty \frac{\epsilon^a}{e^{(\epsilon-\mu)/k_B T} - 1} d\epsilon.$$

Integrating with respect to μ, and using

$$\lim_{\mu \to -\infty} P = 0,$$

we obtain

$$\begin{aligned} P(T,\mu) &= A \int_0^\infty \epsilon^a \left[\int_{-\infty}^{\mu} \frac{d\mu'}{e^{(\epsilon-\mu')/k_B T} - 1} \right] d\epsilon \\ &= -A k_B T \int_0^\infty \epsilon^a \ln \left[1 - e^{(\mu-\epsilon)/k_B T} \right] d\epsilon. \end{aligned}$$

Therefore,

$$\boxed{P(T,\mu) = -A k_B T \int_0^\infty \epsilon^a \ln \left[1 - e^{(\mu-\epsilon)/k_B T} \right] d\epsilon}.$$

3 III

3.1 (7)

The pressure P_{in} is the pressure of the gas trapped inside the balloon, which corresponds to the Helmholtz free energy $F(V)$ of the gas itself. Thus,

$$P_{\text{in}} = -\frac{dF(V)}{dV}.$$

On the other hand, P_{ex} is the external pressure exerted on the balloon. At equilibrium, it balances the effective pressure arising from both the gas and the balloon membrane.

Since the total Helmholtz free energy is

$$F_{\text{tot}}(V) = F(V) + \gamma V^{2/3},$$

the equilibrium condition gives

$$\frac{d}{dV} \left[F_{\text{tot}}(V) + P_{\text{ex}} V \right] = 0.$$

Therefore,

$$P_{\text{ex}} = -\frac{dF_{\text{tot}}}{dV}.$$

Combining the two expressions, we obtain

$$\begin{aligned} \Delta P &= P_{\text{in}} - P_{\text{ex}} \\ &= -\frac{d}{dV} \left[F(V) - F_{\text{tot}}(V) \right] \\ &= \frac{2}{3} \gamma V^{-1/3}. \end{aligned}$$

Thus, the equilibrium pressure difference decreases as the volume of the balloon increases.

4 IV

4.1 (8)

From the result of (7),

$$P_L - P_G = BV^b.$$

Taking the differential,

$$dP_L - dP_G = BbV^{b-1}dV.$$

On the other hand, since the chemical potentials of liquid water and water vapor are equal, their differentials must also be equal:

$$d\mu_L = d\mu_G.$$

Using the Gibbs-Duhem relation at constant temperature,

$$v_L dP_L = v_G dP_G.$$

Combining these equations,

$$v_L(dP_G + BbV^{b-1}dV) = v_G dP_G.$$

Therefore,

$$dP_G = \frac{v_L}{v_G - v_L} BbV^{b-1}dV.$$

Since $v_L \ll v_G$, keeping only the first-order term in v_L/v_G,

$$\boxed{dP_G \approx \frac{v_L}{v_G} BbV^{b-1}dV}.$$

5 IV

5.1 (8)

At equilibrium, the chemical potentials of liquid water and water vapor are equal:

$$\mu_L = \mu_G.$$

Therefore, their differentials are also equal:

$$d\mu_L = d\mu_G.$$

Using the Gibbs-Duhem relation at constant temperature,

$$d\mu_L = v_L dP_L, \qquad d\mu_G = v_G dP_G.$$

Thus,

$$v_L dP_L = v_G dP_G.$$

From (7), we have

$$P_L - P_G = BV^b.$$

Taking the differential,

$$dP_L - dP_G = BbV^{b-1}dV.$$

Combining the above equations,

$$v_L(dP_G + BbV^{b-1}dV) = v_G dP_G.$$

Therefore,

$$dP_G = \frac{v_L}{v_G - v_L} BbV^{b-1}dV.$$

Since $v_L \ll v_G$, keeping only the first-order term in v_L/v_G,

$$\boxed{dP_G \approx \frac{v_L}{v_G} BbV^{b-1}dV}.$$

5.2 (9)

Since v_L is independent of pressure and v_G approximately satisfies the classical ideal gas equation,

$$v_G = \frac{k_B T}{P_G},$$

the differential equation obtained in (8) becomes

$$\frac{dP_G}{P_G} = \frac{v_L Bb}{k_B T} V^{b-1}dV.$$

Integrating both sides,

$$\int_{P_\infty}^{P_G} \frac{dP'_G}{P'_G} = \frac{v_L Bb}{k_B T} \int_{\infty}^{V} V'^{b-1}dV'.$$

Using

$$\lim_{V\to\infty} P_G(V) = P_\infty,$$

we obtain

$$\ln \frac{P_G}{P_\infty} = \frac{v_L B}{k_B T} V^b.$$

Therefore,

$$\boxed{P_G(V) = P_\infty \exp\left(\frac{v_L B}{k_B T} V^b\right)}.$$

Substituting the results of (7),

$$B = \frac{2\gamma}{3}, \qquad b = -\frac{1}{3},$$

we finally obtain

$$\boxed{P_G(V) = P_\infty \exp\left(\frac{2\gamma v_L}{3k_B T V^{1/3}}\right)}.$$

As the droplet volume decreases, the saturation vapor pressure increases.